Negative-Weight Reweighting

The algebra, and why it is an identity

$H \to WW$ + charm · V+jets · arXiv:2510.16217

1 · Where negative weights come from

An NLO cross section splits into real and virtual contributions:

$$\sigma_{\mathrm{NLO}} = \int d\Phi_{n+1}\, R + \int d\Phi_n\, V + \int d\Phi_n\, B$$

R and V are **each divergent**; the divergences cancel only in the sum. To integrate numerically you introduce subtraction terms S:

$$\sigma_{\mathrm{NLO}} = \int d\Phi_{n+1}(R - S) + \int d\Phi_n \left(V + \int_1 S\right) + \int d\Phi_n B$$

Each bracket is now finite — but $R - S$ and $V + \int_1 S$ are **not positive definite**. Events sampled from them carry weights of either sign.

> Negative weights are not a bug or a generator pathology. They are the price of a finite, matched NLO calculation.

2 · Why that hurts

The yield estimator in a bin B:

$$\hat{N}_B = \sum_{i\in B} w_i, \qquad w_i = \pm|w_i|$$

quantity	behaviour
expectation $\langle \hat{N}_B \rangle$	**correct** — the estimator is unbiased
variance $V[\hat{N}_B]$	$\propto \sum_i w_i^2 = \sum_i |w_i|^2$

The **mean** is the signed sum (small), the **variance** follows the unsigned sum (large). Define the effective sample size

$$n_{\text{eff}} = \frac{\left(\sum_i w_i\right)^2}{\sum_i w_i^2} \ll N$$

> In our V+jets, **16.4%** of events had $w < 0$. One DY bin came out 0 ± 41 from $\pm 79{,}000$ weights cancelling — zero content, large error. `autoMCStats` then assigns that bin a huge nuisance.

3 · The decomposition

Any signed density can be written as a **difference of two positive, normalised** densities:

$$p(\vec{x}) \;=\; a\,p_+(\vec{x}) \;-\; b\,p_-(\vec{x})$$

with $a, b \geq 0$ and, since p is itself normalised,

$$\int p = a - b = 1$$

p_+ — the normalised density of **positive-weight** events
p_- — the normalised density of **negative-weight** events
a, b — their total weights, with $a - b = 1$

A Monte Carlo generator samples p_+ with probability $\propto a$ and p_- with probability $\propto b$, assigning $\pm$ accordingly.

4 · The positive-weight probability

At a point $\vec{x}$, the probability that an event drawn there carries $w > 0$:

$$P_+(\vec{x}) \;=\; \frac{a\,p_+(\vec{x})}{a\,p_+(\vec{x}) + b\,p_-(\vec{x})}$$

Write $u(\vec{x}) \equiv a\,p_+(\vec{x}) + b\,p_-(\vec{x})$ — the **unsigned** density.
Then

$$a\,p_+ = P_+\,u, \qquad b\,p_- = (1 - P_+)\,u$$

Substituting into $p = ap_+ - bp_-$:

$$p(\vec{x}) = P_+ u - (1 - P_+)u = \big(2P_+(\vec{x}) - 1\big)\,u(\vec{x})$$

$$\boxed{p(\vec{x}) \;=\; g(\vec{x})\,u(\vec{x}), \qquad g(\vec{x}) \equiv 2P_+(\vec{x}) - 1}$$

5 · Why this solves the problem

$u(\vec{x})$ is the density that $\sum_i |w_i|$ samples — **all events, unsigned**.

So for any observable f:

$$\int f(\vec{x})\, p(\vec{x})\, d\vec{x} \;=\; \int f(\vec{x})\, g(\vec{x})\, u(\vec{x})\, d\vec{x}$$

$$\Longrightarrow \quad \sum_i w_i\, f(\vec{x}_i) \;\widehat{=}\; \sum_i |w_i|\, g(\vec{x}_i)\, f(\vec{x}_i)$$

Both estimate the same distribution. The reweighting is therefore

$$w_i \;\longrightarrow\; |w_i| \cdot g(\vec{x}_i)$$

> **Same yield. Same shape. Smaller variance** — because every event now contributes with the same sign, scaled by $g \in [-1, 1]$, instead of cancelling.

6 · The variance gain, explicitly

	signed	reweighted
estimator	$\sum w_i$	$\sum \lvert w_i \rvert\, g_i$
per-event magnitude	$\lvert w_i \rvert$	$\lvert w_i \rvert \cdot \lvert g_i \rvert \le \lvert w_i \rvert$
cancellation	yes	**no**

Since $|g| \le 1$, every term shrinks *and* nothing cancels. The variance

$$V\left[\sum |w_i| g_i\right] = \sum |w_i|^2 g_i^2 \le \sum w_i^2$$

with equality only where $|g| = 1$, i.e. where the sign is deterministic.

Measured in our sample: $\langle g \rangle = 0.672 = 2(0.836) - 1$, matching the global positive fraction. Range $[-0.991, 0.993]$ — inside $[-1, 1]$, no pathological events.

7 · The one approximation

Everything above is **exact for the true P_+**. In practice P_+ is unknown, so we estimate it with a classifier:

$$\hat{P}_+(\vec{x}) \approx P_+(\vec{x}) \quad \Longrightarrow \quad \hat{g} = 2\hat{P}_+ - 1$$

> **This is the only approximation in the method.** Closure stops being an algebraic guarantee and becomes an **empirical property that must be measured**.

model	`HistGradientBoostingClassifier`, 20-model ensemble
target	$\mathrm{sign}(w)$
features	**20, all generator-level** (`lhe_*`, `genparton_*`)
training	9.8M events
ensemble AUC	**0.829**

8 · Why generator-level features only

$P_+(\vec{x})$ is a property of **the generator**, not of the detector.

The identity $p = g\,u$ holds in the variables the generator actually sampled. If $\vec{x}$ contained reconstruction-level quantities, u would no longer be the density $\sum|w|$ samples, and closure would not follow.

> The paper (§V.2) rejects reco-level inputs for exactly this reason: *"using reconstruction-level variables will not guarantee closure on generator-level variables."*

Our 20 features: `lhe_njets`, `lhe_nb/nc/nuds/nglu`, `lhe_ht`, `lhe_vpt`, `lhe_alphas`, `genparton_multiplicity`, `genparton_n_pt20/100/200`, incoming parton PDG IDs, and the two leading partons' p_T and η.

9 · Train loose, infer tight

The signal region has far too few V+jets events to train on — that starvation is the problem being solved.

	region
train	all base cuts **+ veto of the eμ SR topology**
infer	the tight eμ signal region

Disjoint by construction. An event that is not "exactly one eμ pair" can never be an SR event, so training and inference sets cannot overlap — no event-ID bookkeeping needed, and no overfitting bias.

Disjoint **events**, but *not* an orthogonal **phase space**. g must be interpolated over the SR's support, never extrapolated — so an anti-SR training region (e.g. c-depleted) would be actively harmful.

10 · Validation

check	result
training-region closure	**0.994** on 9.8M events
calibration of $\hat{P}_+$ in the SR	on the diagonal, $P_+ \approx 0.15 \to 0.93$
ensemble spread δg	mean **0.006**, max 0.467
SR closure, **before** renorm	**1.058** (a 5.8% overshoot)

The SR is a small, kinematically-biased corner of the training domain, so the per-event g does not average to the *local* positive fraction:

dataset	rows	$\sum \lvert w \rvert g \,/\, \sum w$
`DYto2L_2Jets_50`	9,646	1.014
`WtoLNu_2Jets`	485	1.133
total	10,205	**1.058**

10b · What we do about it

The reweighted V+jets template is renormalised to the nominal yield, per dataset.
Applied factors: DY×0.986, WtoLNu×0.900.

So the yield is **restored exactly** — after renormalisation the reweighted and nominal templates have the same integral by construction.

	preserved?
yield	**yes** — enforced by the renorm
shape	**yes** — the renorm is a single scale factor per dataset
variance reduction	**yes** — n_{eff} comes from per-bin variance, untouched by a global scale

This renormalisation is our extension, not the paper's — their evaluation region was not this starved, so their closure was clean without it. `CMS_negrw_vjets` covers the residual **method** uncertainty (the ensemble spread), not the 5.8% offset, which is removed rather than assigned an uncertainty.

11 · What enters the analysis

```
# per event, dumped alongside the nominal weight
weight_negrw     = 2 * P_plus_ensemble_mean(x) - 1      #  = g
weight_negrw_std = 2 * P_plus_ensemble_std(x)           #  = delta g
```

column	role
`weight_negrw`	the reweight factor g applied to $\lvert w \rvert$
`weight_negrw_std`	the 20-model spread → shape nuisance **`CMS_negrw_vjets`**

Applied only to the V+jets samples the ensemble was trained for, via an **anchored** dataset match — a substring gate would also catch `WplusH_WtoLNu_...` and silently reweight a Higgs template with a V+jets model.

> `CMS_negrw_vjets` profiles to **0.0%** — the method's own uncertainty is negligible.

Summary

$$p(\vec{x}) = \big(2P_+(\vec{x}) - 1\big)\, u(\vec{x})$$

An algebraic identity, not an approximation.

Same yield, same shape, no cancellation.

The only approximation is $\hat{P}_+$ from a finite classifier —
and its residual is a nuisance that profiles to zero.